Floquet-Bloch Theory

In our research we often do not only study arrangements of finitely many resonators but also cases in which there are an infinite number of them. Most often, we assume the resonators to be arranged periodically,

i) chains ii) sheets iii) crystals

so that they essentially form a lattice. The reason for this being - perhaps somewhat counterintuitively - that such infinite systems are easier to study due to a lack of boundaries which may introduce edge effects.

In what follows, we will familiarise ourselves with one of the key techniques for studying periodic structures, Floquet-Bloch theory. As crystals are ubiquitous in nature, these techniques find application in many physics and non-physics related fields beyond our research. For the sake of simplicity, we will study the Schrödinger operator

$$A(x) = -\Delta + V(x)$$

where we assume $V(x) \in C^\infty(\mathbb{R}^n)$ to be 1-periodic i.e.

$$V(x + e_i) = V(x)$$

Considering A as an operator

$$H^{-1} \subset H^1 \longrightarrow H^{-1}, \quad u \longmapsto \varphi_u(w) = \int \nabla u \nabla \bar{w} + V u \bar{w} \, dx$$

we are interested in the spectrum of A, i.e. the values λ s.t. $(A-\lambda)$ is not invertible.

Denote T_i the unit shift in the ith coordinate, i.e.

$$T_i u(x) = u(x+e_i).$$

Note that due to the periodicity of V, A and T_i commute:

$$\begin{aligned} A T_i u &= -\Delta u(x+e_i) + V(x) u(x+e_i) \\ &= -\Delta u(x+e_i) + V(x+e_i) u(x+e_i) \\ &= T_i A u \end{aligned}$$

Recall from linear algebra that commuting operators have a common basis of eigenfunctions (of course, working in infinite-dim. vector spaces this is a bit more complicated but morally true)
What then are the eigenfunctions of T_i? Well, suppose (λ, u) satisfies

$$\begin{aligned} T_i u = \lambda u &\iff u(x+e_i) = \lambda_i u(x) \\ &\impliedby u(x+e_i \cdot n) = \lambda_i^n u(x) \end{aligned}$$

→ It is enough to know u in $[0,1)^n$! Moreover, in most reasonable settings we want u to be bounded, which implies $|\lambda| = 1$.
Write $\lambda_j = e^{ik_j}$ for $k_j \in [0, 2\pi)$ and consider

$$p(x) := e^{-i\langle k, x\rangle} u(x)$$

This function must be periodic! Hence we should look for solutions

$$u(x) = e^{i\langle k, x\rangle} p(x)$$

where $k \in [0, 2\pi)^n$.

Unfortunately, this is all a bit handwavy. In particular, we consider A as an operator (on a subspace of) H^{-2}, so it is not at all clear why this should work.

Lattices & other preliminaries

We will first need to settle some notation and definitions

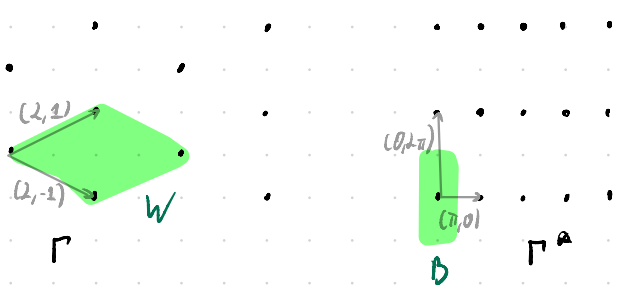
Definition: A lattice $\Gamma \subset \mathbb{R}^n$ is the set of all integer combinations of some n linearly independent vectors v_i.

$$\Gamma = \{ \gamma \in \mathbb{R}^n \mid \gamma = \sum a_i v_i \,,\ a_i \in \mathbb{Z} \}$$

Definition: The dual lattice $\Gamma^* \subset \mathbb{R}^n$ of a lattice Γ is defined as follows:

$$\Gamma^* = \{ k \in \mathbb{R}^n \mid \langle k, \gamma \rangle \in 2\pi\mathbb{Z} \ \ \forall \gamma \in \Gamma \}$$

Example:



Moreover, we define the **fundamental cell** W as a closed subset of $\mathbb{R}^n$ s.t. its Γ shifts intersect with Γ only at the boundary.
Similarily, we define the **Brillouin zone** B as the **Voronoi cell** of Γ^*, i.e. the closure of the set of all points s.t. the origin is the closest lattice point of Γ^*.

Finally, we define the tori

$$\mathbb{T} = \mathbb{R}^n / \Gamma \quad \text{and} \quad \mathbb{T}^* = \mathbb{R}^n / \Gamma^*$$

We will consider $L^2(\mathbb{T})$, $L^2(\mathbb{T}^*)$ as the measure spaces (**Haar measure**) associated with the unique, normalised, translation invariant measure

Let $\mathcal{H}$ be a Hilbert space and $f \in L^2(\Pi, \mathcal{H})$. Define

$$\mathcal{F}[f](k) := \int_{\Pi} f(x)\, e^{-ikx}\, dx$$

Theorem: $\mathcal{F} : L^2(\Pi, \mathcal{H}) \longrightarrow \ell^2(\Gamma^*, \mathcal{H})$ is an isometry

Back to Floquet-Bloch Theory

To rigorously justify our initial arguments, we will make use of a transformation somewhat reminiscent of the Fourier-transform. First, denote

$$f_\gamma(x) := f(x-\gamma)$$

Definition: The Floquet-transform is the map defined by

$$\begin{aligned} U_\Gamma : L^2(\mathbb{R}^n) &\longrightarrow L^2(B, L^2(W)) \\ f &\longrightarrow U_\Gamma[f](x,k) := \sum_{\gamma \in \Gamma} f_\gamma(x)\, e^{ik\gamma} \end{aligned}$$

Proposition: The Floquet-transform satisfies the following properties

i) $U_\Gamma[f](x+\gamma, k) = U_\Gamma[f](x,k)\, e^{ik\gamma}$, $\quad \gamma \in \Gamma$

ii) $U_\Gamma[f](x, k+k') = U_\Gamma[f](x,k)$, $\quad k \in \Gamma^*$

Even more importantly is the fact that U_Γ is invertible:

$$f(x) = \int_{\Pi^*} U_\Gamma[f](x,k)\, dk$$

Finally, one can show that the Floquet-Bloch transform is actually an isometry. Consider the map

$$f \in L^2(\mathbb{R}^n) \longleftrightarrow \{f_\gamma\}_{\gamma} \in \ell^2(\Gamma, L^2(W))$$

this is clearly an isometry. The Fourier-transform then maps

$$\ell^2(\Gamma, L^2(W)) \longleftrightarrow L^2(\Pi^*, L^2(W))$$

Thus we have proven

Theorem $U_\Gamma : L^2(\mathbb{R}^n) \longrightarrow L^2(\Pi^*, L^2(W))$ defined by

$$U_\Gamma[f](x,k) = \sum_{\gamma\in\Gamma} f_\gamma(x)\, e^{ik\gamma}$$

with inverse

$$U_\Gamma^{-1}[F](x) = \int_{\Pi^*} F(x,k)\, dk$$

is an isometry.

Lets go back to our motivation of understanding the spectrum of

$$A(x) := -\Delta + V(x)$$

where we know locally $V \in C^\infty(\Pi)$. Note that $A(x)$ commutes with the Floquet transform

$$\begin{aligned} U[Af](x,k) &= \sum (Af)_\gamma(x)\, e^{ik\gamma} \\ &= \sum \left(-\Delta f|_{W-\gamma}(x-\gamma) + V|_{W-\gamma}(x-\gamma)\, f|_{W-\gamma}(x-\gamma) \right) e^{ik\gamma} \\ &= \sum A f_\gamma(x)\, e^{ik\gamma} \\ &= A\, U_\Gamma[f](x,k) \end{aligned}$$

However, our operator A requires some sort of differentiability, so just understanding how $L^2(\mathbb{R}^n)$ transforms is not enough. To make matters worse, it does not hold that

$$U_\Gamma[H^2] = L^2(\Pi^*, H^2(W))$$

we only have

$$U_\Gamma[H^1] \subsetneq L^2(\mathbb{T}^n, H^1(W))$$

For instance, let $n=1$ and consider

$$\mathcal{F}(x,k) := \sum 1_W(x-\gamma)\, e^{ik\gamma} = 1$$

Then

$$U_\Gamma^{-1}[\mathcal{F}] = 1_W \notin H^1(\mathbb{R}) \subset L^2(\mathbb{R})$$

But clearly $\mathcal{F}(x,k) \in L^2(\mathbb{T}^n, H^1(W))$

Define $E := \mathbb{R}^n \times \mathbb{C}$ and consider the action

$$\gamma[k,\gamma](x,c) = (x+\gamma, e^{ik\gamma}c)$$

We define

$$E_k := E/\gamma[k,\gamma]$$

Denote further

$$H^1_k(W) := \{ f \in H^1_{loc}(\mathbb{R}^n) \mid f(x+\gamma) = f(x)\, e^{ik\gamma} \ \forall \gamma \in \Gamma \}$$

Then the map

$$\sigma_k : H^1_k(W) \times \mathbb{R}^n \longrightarrow E_k, \quad \sigma_k[f](x) = (x, f(x))$$

defines a section of E_k. Hence there is a natural identification between functions

$$f \in H^1_k(W) \quad \sim \quad \text{sections of } E_k.$$

Moreover, it is easy to see that for $f \in H^2(\mathbb{R}^n)$

$$U_\Gamma[f](\cdot, k) \in H^2_k(W)$$

Hence U_Γ maps $H^2(\mathbb{R}^n)$ to $\bigcup_k H^2_k(W)$.
This now properly justifies our initial intuition regarding the eigenfunctions of A. However, there is another useful identification:

Consider now the map

$$\phi_k : H^2_k(W) \longrightarrow H^2(\mathbb{T}), \quad f \longrightarrow e^{-ikx} f$$

This again is an isometry. We thus define

$$\begin{aligned} A(k) : H^2(\mathbb{T}) &\longrightarrow H^2(\mathbb{T}) \\ f &\longrightarrow \phi_k \circ A \circ \phi_k^{-1}[f] = e^{-ikx} \circ A \circ e^{ikx} f \end{aligned}$$

To put in words, $A(k)$ acts on $H^2(\mathbb{T})$, as A acts on $H^2_k(W)$. The advantage of studying $A(k)$ as opposed to A is that $A(k)$ acts on a compact manifold. Since A is elliptic, its inverse is compact and therefore its spectrum is discrete. We have

$$\sigma(A) = \bigcup_k \sigma(A(k))$$

Note that formally, there is a difference between
$A(k) : H^2(\mathbb{T}) \to H^{-2}(\mathbb{T})$
$A : H^2_k(\mathbb{R}) \to (H^2_k(\mathbb{R}))^*$
this is rotating operator vs rotating domain.

Example 1: We take

$$\begin{aligned} A : H^2(\mathbb{R}) \subset H^{-2}(\mathbb{R}) &\longrightarrow H^{-2}(\mathbb{R}) \\ Af &= -\frac{d^2}{dx^2} f \end{aligned}$$

i.e. A is just the Laplacian. Trivially, A is 1-periodic. We now that

$$\sigma(A) = [0, \infty).$$

In this case it is easy to compute $A(k)$:

$$A(k) = e^{-ikx} \circ A \circ e^{ikx} = -\frac{d^2}{dx^2} - 2ik\frac{d}{dx} + k^2$$

Observe that the eigenfunctions of $A(k)$ are just $f_n(x) = \exp(i2\pi n x)$:

$$A(k) f_n(x) = (4\pi^2 n^2 + 4\pi k n + k^2) f_n(x)$$

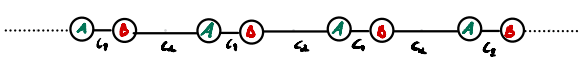
we immediately see that

$$\lambda_n(k) = (2\pi n + k)^2$$

Varying k from 0 to 2π, we immediately get

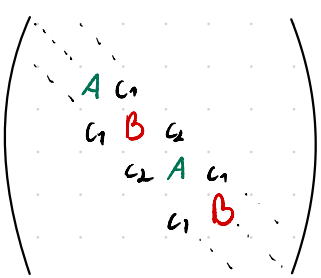
$$\sigma(A) = \bigcup_k \sigma(A(k))$$

Example 2: Consider a dimer chain



with on-site potentials A & B and intra-cell interactions c_1, c_2. The Hamiltonian of this system is given by the Laurent operator

$L :=$



This system is again periodic, if we define the shift operator

$$S : \ell^2(\mathbb{Z}, \mathbb{C}^2) \longrightarrow \ell^2(\mathbb{Z}, \mathbb{C}^2) \quad (a_n, b_n) \longrightarrow (a_{n+1}, b_{n+1})$$

Then $S^{-1} \mathcal{L} S = \mathcal{L}$. Since we already are on a lattice, the Floquet-transform is just the Fourier-transform:

$$\{(a_n, b_n)\}_n \in L^2(\mathbb{Z}, \mathbb{Z}^2) \longrightarrow \sum_{n \in \mathbb{Z}} \begin{pmatrix} a_n \\ b_n \end{pmatrix} e^{ikn}$$

Thus, our operator $\mathcal{L}$ turns into

$$\mathcal{L}(k) := \begin{pmatrix} 0 & c_2 \\ 0 & 0 \end{pmatrix} e^{ik} + \begin{pmatrix} A & c_1 \\ c_1 & B \end{pmatrix} + \begin{pmatrix} 0 & 0 \\ c_2 & 0 \end{pmatrix} e^{-ik}$$

The eigenvalues now satisfy

$$\det(\mathcal{L}(k) - \lambda(k)\mathbb{1}) = 0 \Leftrightarrow \lambda^2 - \lambda(A+B) + AB - 2c_1c_2\cos k - c_1^2 - c_2^2$$

Hence

$$\lambda_\pm(k) = \frac{A+B \pm \sqrt{(A+B)^2 - 4(AB - 2c_1c_2\cos k - c_1^2 - c_2^2)}}{2}$$

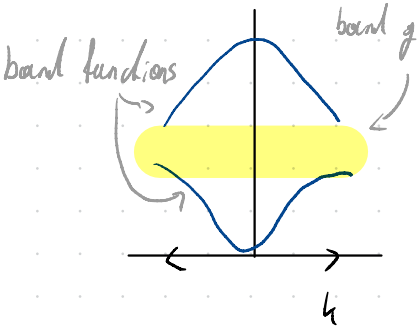
Through these two examples we noticed another interesting feature: The eigenvalues of $A(k)$ are continuous functions of k! Indeed one can go even further that they are actually piecewise analytic!

Definition: The function $\lambda_j(k)$ is called the j-th band function

Definition: The dispersion relation is the set

$$B_H := \{(k, \lambda) \in \mathbb{R}^{n+1} \mid \lambda \in \sigma(H(k))\}$$

In 1D the dispersion takes a particularly nice form:



i) If $A(\cdot)$ is real and self-adjoint, it is reflection symmetric.

ii) Bands can touch but they can never overlap

iii) The band functions are monotone

Proof: For i) suppose $u(x) = e^{ikx} p(x)$ with p periodic is a Bloch eigenfunction. Since A is real and s.a., we have

$$\begin{aligned}(A-\lambda)\, e^{ikx} p(x) &= 0 \\ &= \overline{(A-\lambda) e^{ikx} p(x)} \\ &= (A-\lambda) e^{-ikx} \overline{p(x)}\end{aligned}$$

ii) & iii) are a straightforward application of Picard-Lindelöf: That is, an ode has two linearly independent solutions. If the bands where to overlap or not monotone, there would exists some $\lambda \in \mathbb{R}$ s.t. $\lambda \in \sigma(h)$ for more than two different values of h!

Sources & Further Reading

i) An Overview of Periodic, Elliptic Operators, Peter Kuchment

ii) Functional Analysis, Spectral Theory and Applications, Manfred Einsiedler, Thomas Ward